

Yuhan LIU

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PROFESSIONAL SUMMARY

Ph.D. student with hands-on experience in hardware prototyping, SolidWorks-based mechanical design and CAD modeling, experimental testing, and quantitative performance validation across RF metamaterials and wearable haptic interfaces, with prior industry internship experience in optical lens fatigue/stray-light testing and fluidic 3D-printing system integration. Experienced in cross-functional technical communication, human-machine interface development, prototype integration, test workflow development, and parameter optimization.

EDUCATION AND TRAINING

Ph.D. in Mechanical Engineering	Boston University, Boston, MA	Present
M.S. in Electrical Engineering	Tsinghua University, China	May 2023
B.S. in Mechanical Engineering	Xi'an Technological University, China	Jul 2019

RESEARCH EXPERIENCE

Wireless RF Metamaterial Engineering for MRI Signal Amplification

Doctoral student Researcher | Laboratory for Microsystems Technology, Boston University, MA | Sep 2024 – Present

- Designed a wireless volumetric RF metamaterial device operating at 127.74 MHz, achieving 2.24× SNR enhancement, reducing signal variation from 54.8 % to 12.1 % compared to advanced commercial multi-channel coils, while reducing system costs, tested in a Philips 3T MR scanner.
- Built and validated RF hardware prototypes through electromagnetic simulation, structural design, PCB fabrication, RF characterization, and clinical MRI testing; developed quantitative benchmarking workflows for signal enhancement and imaging performance evaluation.

Wearable Haptic Interfaces and Sensors for Human-Machine Interaction

Graduate Researcher | Microelectromechanical Systems Lab, Tsinghua University, China | Sep 2020 – May 2024

- Designed multilayer PDMS-based elastomer composites with tunable stiffness for soft haptic actuation, reducing the driving voltage from 500 V to 50 V (~1 mN/V efficiency).
- Integrated soft materials, actuator arrays, and MATLAB/Arduino-based control to develop programmable wearable haptic interfaces capable of spatially and temporally tunable mechanical stimulation.
- Developed an emotion- and direction-based haptic mapping framework using decoupling-coupling analysis, translating perceptual cues into quantitative actuation parameters and validating performance with 20+ participants, achieving 64.6% untrained and 95.8% trained recognition accuracy.
- Contributed to wearable sensing and haptic-interface projects for accessible HCI applications, including music perception and cultural-heritage interaction, supporting device prototyping, signal processing, and user-centered evaluation.

SELECTED FIRST-AUTHOR PUBLICATIONS

- [1] **Yuhan Liu**, et al. Circularly Polarized Metamaterial Cage for Homogeneous Signal-to-Noise Ratio Enhancement in Magnetic Resonance Imaging. *Advanced Materials*. 2026, e16569. [DOI/10.1002/adma.202516569](https://doi.org/10.1002/adma.202516569) (IF:27.4)
- [2] **Yuhan Liu**, et al. Emotional and Directional Enabled Programmable Flexible Haptic Interface for Enhanced Cognition in Disabled Community. *Research*. 2025, 8, 0714. [DOI/10.34133/research.0714](https://doi.org/10.34133/research.0714) (IF: 10.9)
- [3] **Yuhan Liu**. Advanced Mental Health Monitoring with Modular Pixel Hydrogel Skin---Real-time Emotional and Stress State Management, *MRS Fall Meeting*, Poster Presentation, Boston, USA, 2024.

AWARDS & LEADERSHIP

10/2022	Vice President, Tsinghua SIGS Student Union
10/2022	Outstanding Student Leaders, Tsinghua University
10/2021	First Prize Academic Excellence, Tsinghua Shenzhen International Graduate School (top 10%)
12/2020	National Gold of the 4th Youth Innovation and Entrepreneurship Competition, Youth Innovation and Entrepreneurship Competition Organizing Committee
09/2018	First Prize Academic Excellence, Xi'an Technological University (top 0.5%)

SKILLS

Hardware Design & Prototyping:	SolidWorks, 3D printing, laser cutting, PCB fabrication, silicone molding, soft-material fabrication
Testing & Characterization:	Mechanical/electrical characterization, RF characterization, VNA testing, experimental validation, performance benchmarking
Software & Analysis:	CST Microwave Studio, COMSOL, MATLAB, Python, signal processing
Electronics & Control:	Arduino, LabVIEW, actuator arrays, embedded control basics